

Edge intelligence-driven digital twin of CNC system: Architecture and deployment

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ABSTRACT

In recent years, digital twin (DT) technology has gradually become the primary way to achieve the intelligence of CNC systems. However, with the development of next-generation information technologies such as artificial intelligence (AI) and its wide application in CNC systems, the limitation of computing power and network resources has become one of the urgent problems that must be solved by the DT of CNC systems. To address these problems, a theoretical modeling method for CNC systems based on its hierarchical structure is proposed first, and the edge intelligence (EI) technology is introduced to support the deployment of DT models. Meanwhile, a model partitioning method and a model selection algorithm are proposed to support real-time model response in the model deployment process. In addition, an application case of EI-driven DT of CNC system is given to diagnose and predict the tool wear during machining processes.

1. Introduction

The rapid development of next-generation information technology and its deep integration into advanced manufacturing technologies drive the transformation of traditional manufacturing industries to intelligent manufacturing [1–3]. As an enabling technology for intelligent manufacturing, digital twin (DT) technology [4] is widely used in CNC machine tools. Currently, the research on the DT of CNC machine tools is mainly focused on data acquisition and simulation analysis [5–9]. There are still some difficulties in bridging the gap between modeling, implementation, and deployment of the DT model. As the brain that guides the machining of CNC machine tools [10], the CNC system is an indispensable carrier of the machine tools' DT system. To realize ubiquitous sensing and intelligent control of the machining process, research on key technologies of the DT of CNC systems is conducted.

The CNC system is a complex system composed of various subsystems, which is influenced by numerous factors in machining processes. In this condition, artificial intelligence (AI) algorithms represented by machine learning [11] are required to realize the DT of CNC systems. The rapid development of AI algorithms provides more solutions for DT, but also puts forward higher requirements including

real-time synchronizing, accurate mapping, and high-fidelity. Existing CNC systems have limited computing power to analyze and process massive amounts of data, resulting in many enterprises tending to upload all data collected from equipment to remote cloud datacenters [12–14]. However, the shop floor network resources are restricted, this cloud-dependent data processing will cause higher transmission costs, network congestion, data delivery lag, and inadequate privacy [15,16].

Compared with pure cloud computing, edge computing is suitable for scenarios such as real-time processing, short-period communication, and local decision-making [17–19]. It distributes the combinable CNC DT model to the edge of the network close to the user by providing a distributed computing platform [20–22], which can effectively solve these problems. In order to speed up the intelligent process of CNC system, this paper deploys Deep Neural Networks (DNN) models through edge intelligence (EI) to improve the performance of AI applications in the DT of CNC system. The contributions of this paper can be summarized as follows.

- 1) Propose a theoretical modeling approach based on the hierarchical structure of CNC systems and introduce EI technology to deploy DT models. A DT architecture for CNC systems driven by EI is proposed, and a generic DT modeling approach is given.

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- 2) Propose a new model partitioning method in the deployment of DT models utilizing the distributed characteristics of edge computing to improve the throughput rate while meeting the real-time requirements of DT applications.
- 3) Proposes an adaptive model selection algorithm that automatically selects the algorithm model according to different application requirements for real-time and system accuracy.
- 4) An application case of EI-driven DT of CNC system is given for diagnosing and predicting tool wear during the machine tool machining process.

The overall structure of the study takes the form of six sections, including this introductory section. Section 2 provides a comprehensive review of the related works in the field of DT and EI. Section 3 introduces the EI-driven DT of CNC system. Section 4 describes how to partition and deploy intelligent algorithms via edge computing. Section 5 presents a case study of EI-driven DT of CNC system. Conclusion and potential future directions are summarized in Section 6.

2. Related works

This section reviews the related research on DT and EI, summarizes the current situation of development of them, and illustrates the research gap to outline the background and explain the significance and importance of the research.

2.1. Digital twin

The DT [5] is a digital model of an existing or future physical entity object that senses, diagnoses, and predicts the state of the physical entity object in real time through actual measurements, simulation and data analysis, regulating the behavior of the physical entity objects, evolving itself through mutual learning between related digital models, and simultaneously improving stakeholders' decisions during the life cycle of physical entity objects. With the rapid development of industrial intelligence, DT shows promising application prospects in intelligent manufacturing. Tao et al. [23] proposed a new DT-led product design, manufacturing, and service approach and gave DT's application method and framework in these three stages. Liu et al. [24] proposed a DT-driven manufacturing system for automated flow operation with a rapid, personalized design approach and discussed how DT could be applied to define and optimize system behavior. Lu et al. [25] discussed the connotation, application scenarios, and research issues of DT-driven intelligent manufacturing in the context of Industry 4.0. Wang et al. [26] established a DT model for the shop floor and proposed a proactive material handling method for the shop-floor to improve the productivity of the shop floor.

Many leading Industrial Enterprises have also developed various applications under the guidance of the DT concept. General Electric (GE) [27] built asset, system, and cluster-level DT based on the Predix platform to understand product health and predict asset performance. Microsoft [28] improved its DT portfolio to build next-generation Internet of Things (IoT) solutions that simulate the real world. Siemens [29] used DT to create an integrated enterprise model of a manufacturing process production system that digitizes the entire production process from design and production to operations.

Both academia and industry have introduced their own DT concepts and clarified the development prospects of DT. However, the technical aspects of DT establishment are not well-established. Moreover, the current research focuses mainly on the application of DT to solve product design, operation and maintenance problems [6], without considering the application of DT in unit-level equipment such as CNC system, ignoring the vital role of DT technology in the shop floor.

2.2. Edge intelligence

Edge computing reduces the pressure on the cloud by sinking computation and services closer to the data source and improves the efficiency of DT applications. Nowadays, the contradiction between huge computational demand and limited computing resources in industrial sites is increasingly prominent. To solve this problem, Zhou et al. [21] proposed the concept of EI, which deploys intelligent algorithms close to users and the edge of the network. Based on this, a series of deep learning model training and inference optimization techniques for EI has emerged in academia to solve over-reliance on hardware resources intelligent algorithms represented by deep learning. Lane et al. [30] used model pruning to compress DNN models, thus reducing model complexity and resource requirements to better achieve low-latency model inference on resource-constrained devices. Surat et al. [31] proposed a distributed deep neural network framework that accelerates deep learning model inference by processing the output results of the more advanced network layers to end model inference early and obtain the final output results. Kang [32] designed a lightweight scheduler *Neurosurgeon*, which automates split computation between devices and datacenters at the granularity of neural network layers to achieve collaborative model inference at the edge for best latency. Zhao et al. [33] proposed a distributed edge computing model based on federated learning, which uses lightweight neural networks to reduce model training time and relieve inference pressure on end devices. Li et al. [34] proposed *Edgent*, a deep learning inference acceleration framework based on end-edge collaboration, combining two technical means, including model segmentation and model early exit, which meet the application requirements by optimizing the selection of model exit points and segmentation points.

Although existing works managed to solve the deployment and inference problems of DNN in resource-constrained environments, there are still accuracy sacrifices caused by model pruning, early exit, and light-weighting of DNN models to some extent. At the same time, collaboration computing primarily uses serial execution, which is challenging to meet the real-time requirements of industrial production. In the case of high frequency of data, it also brings more excellent storage and computing pressure to the end devices.

Zhang et al. [6] proposed the concept of cyber physical machine tools (CPMT) to apply edge computing to DT of machine tools. It reduces the mapping latency and improves the computing efficiency for DT applications. Nevertheless, they did not give an explicitly modeling method for the DT of CNC system. Furthermore, they only applied edge computing in the stage of data fusion. The calculation of complex models still needs to be uploaded to the cloud for processing, unable to maximize the utilization of computing resources at the edge fully.

In response to the strengths and weaknesses of the above researches, Section 3 of this paper proposes a DT architecture for CNC system driven by edge intelligence, which can further enhance the real-time information interaction between physical entities and DTs and strengthen the application of DT in actual production.

3. EI-driven DT of CNC system

According to the development needs of intelligent manufacturing, a theoretical method of constructing the CNC system pattern is proposed in this section. EI technology is introduced to propose an EI-driven CNC system DT architecture to provide solutions for realizing DT of CNC system. The DT modeling process of CNC systems and the deployment of intelligent algorithms via edge computing are also presented.

3.1. Intelligent needs

CNC system is a kind of mechatronics equipment, the main functions of which are interpolation operation and servo control. It is widely used to process complex, multi-variety, precise, and small-batch parts [35,

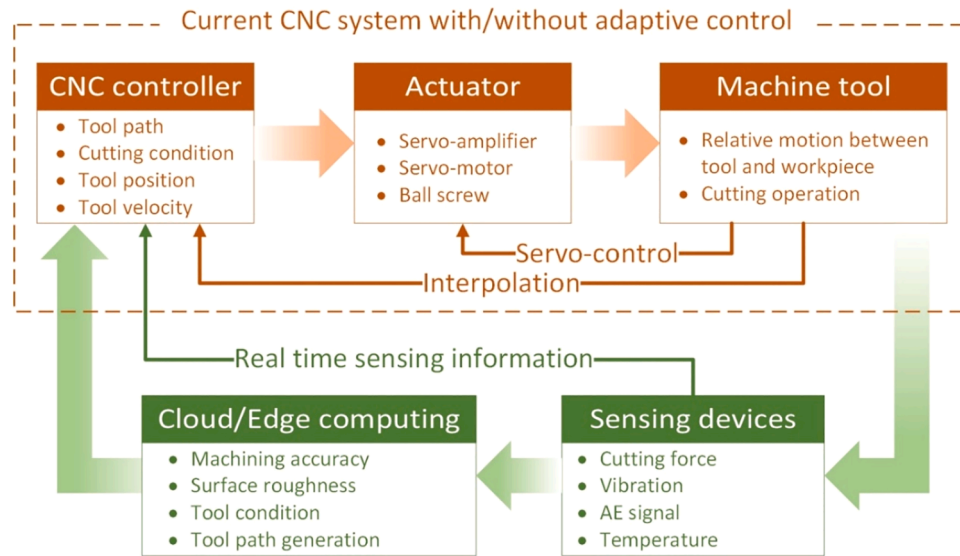


Fig. 1. Intelligent CNC System with adaptive control.

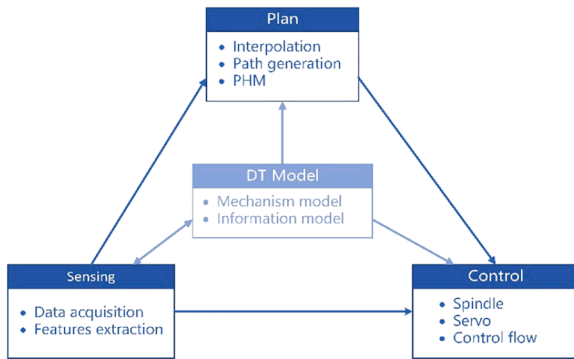


Fig. 2. Sensing-Plan-Control-Modelling pattern for CNC System.

[36]. The development of information and communication has promoted the CNC system from the traditional closed structure to openness, increasing its usage in manufacturing workshops. With the advent of the Internet of Everything (IoE) era, the rapid progress of technologies such as the IoT, AI, and edge computing have created conditions for the development of intelligent CNC systems. Fig. 1 shows the manufacturing closed loop of the intelligent CNC system. The introduction of sensing and computing devices brings adaptive control capabilities such as self-awareness, self-adjustment, self-prediction, and self-assessment to existing CNC systems [37], enriching the functions and services of CNC systems and extending their structure. At the same time, it also creates new requirements for the system.

- 1) The development trend of intelligent manufacturing makes a claim on breaking data silos and realizing the interconnection of CNC equipment.
- 2) The massive amount of data brought by the IoE and the increasing number of delay-sensitive applications have put forward more stringent requirements for system real-time.
- 3) Industrial producers in fields such as aerospace are reluctant to upload all data to the cloud, considering data privacy.
- 4) The growing tension between limited resources at industrial sites and the production goals of accuracy, quality, and efficiency has created new requirements for balancing system throughput and accuracy.

3.2. Modeling pattern and architecture

The DT of CNC system is a necessary means to achieve the intelligence of CNC system. As a complex motion control system, the CNC system is composed of functional components at different levels. Therefore, the DT model of the CNC system can also be constructed with the DT models of its components. Each sub-component can be abstracted as three functional components sharing one DT model, as shown in Fig. 2.

- **Sensing.** This sub-component measures the actual motion performed and received information extracted from sensors to provide data support for updates of the DT model.
- **Plan.** This sub-component makes work plans based on requirements and constraints, such as interpolation, path generator, and Prognostics and Health Management (PHM).
- **Control.** This sub-component, combining the sensing and decision results, issues control commands to the actuator under the guidance of the DT model.
- **DT model.** It consists of two parts: Information Model (IM) and Mechanism Model (MM), in charge of managing the details of the above three functional sub-components. Among them, the IM contains the composition of the information it is concerned with and the rules for describing the information [25], and the MM is able to reflect the physical entity's behavior as well as the rules. Detailed information will be presented later.

With the increase in the variety and quantity of sensors in industrial sites, the data volume explodes, and the scale of IMs consequently becomes larger. Meanwhile, the development of AI technology improves diversity of machine learning algorithms, which leads to an increase in the complexity of MMs. To meet the considerable demand for computing resources, Liu et al. [38] proposed a cloud-based machine tool DT architecture, which provides guidelines for the development and implementation of the existing DT of the CNC system. In the physical layer, the architecture collects real-time manufacturing data through CNC controllers and different types of data acquisition devices, then uploads it to the IM through the network layer. In the cyber layer, the ultra-high computing capacity of the cloud datacenter combines real-time data with intelligent algorithms to simulate and optimize the machining process of CNC machine tools and provide value-added services to users.

However, some key functions of CNC systems, such as error compensation and trajectory planning, have strict time limits. This

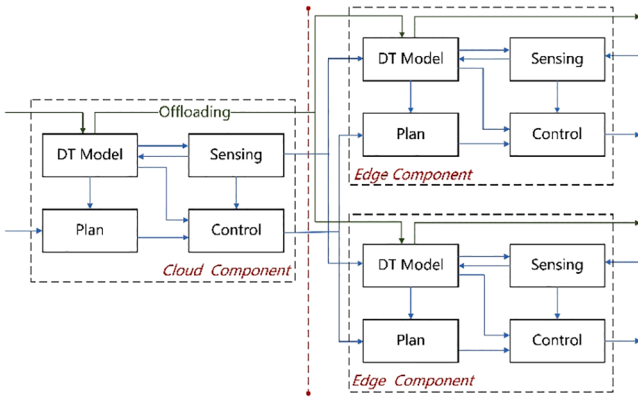


Fig. 3. Edge computing-based hierarchical component model.

cloud-dependent data processing method introduces uncertain time latency due to bandwidth and other factors. As a result, the outcomes of real-time data processing cannot be fed back to the physical machine on time. Therefore, the conventional DT architecture of the CNC system cannot guarantee the consistency and synchronization of DT data interactions, which can lead to the isolation of the physical machine tool from the DT model.

The key advantage of EI is that it can provide users with low-latency and high-reliability computing services through distributed collaborative computing. Moreover, the characteristics of the hierarchical structure of CNC systems enable their DT models to be deployed in a distributed manner. Therefore, edge computing is introduced into the DT of CNC systems.

In this paper, according to the requirements of CNC system tasks, the functions with complex computation and low real-time requirements are abstracted as cloud components, while the functions with high real-time requirements are abstracted as edge components. As shown in Fig. 3, the edge components collect sensing information and upload it to the cloud. In contrast, the cloud components divide the computational tasks according to the real-time requirements of the functions and offload them to appropriate edge components for data processing. Based on the above, this paper proposes an EI-driven DT architecture for CNC systems. As shown in Fig. 4, which includes three layers.

- **Cyber layer.** This layer consists of hierarchical cloud components and edge components together. This paper analyzes the real-time machining data by embedding intelligent algorithms such as process optimization, machining simulation, and health management to

build a MM of the CNC system, which forms the complete DT model together with the IM. Finally, the DT model interacts with the users via digital threads and stores the relevant data processing results and knowledge in the DT model and database to support subsequent application services.

- **Physical layer.** This layer contains the CNC system and data acquisition devices. The data acquisition devices provide a data source for *Sensing*, which captures and uploads the real-time data to the edge and cloud for processing. The CNC system instructions from *Control* to regulate and control the manufacturing process.
- **Application layer.** This layer provides the user with the interface to the subcomponents. It provides intelligent services based on the different algorithms deployed in the cyber layer and the various working requirements faced by the CNC system, such as real-time monitoring of machine status, fault diagnosis and prediction, to achieve intelligent control of the production process.

3.3. DT model

The existing architecture of CNC system focuses only on machining control. DT introduces the machining process and the machining result, thus enriching the CNC system architecture. The DT collects data from physical entities through digital threads, builds IMs and MMs that reflect the system's geometric, physical, behavior and rule-based characteristics, and performs integration and fusion of the data and models. CNCs need to be pre-abstracted into a pre-defined IM during the creation of the DT model, which contains the components and the rules for describing the information [25]. The DT modeling process for CNC system is illustrated in Fig. 5.

Information model. During the machining process, the CNC controller and data acquisition devices capture real-time data generated by the machining process and then transmit the data to peripheral devices for processing via Industrial Ethernet, Fieldbus, 4G/5G, RFID, Wi-Fi, or Bluetooth, thus linking several types of data to the CNC. The CNC system is a complex system composed of a variety of subsystems/components. As a result, the DT of the CNC system is also composed of DT models of each corresponding subsystem/component. In previous studies, we specified the structure of the basic IM of CNC equipment by analyzing the characteristics of the hierarchical structure of CNC equipment [39] and described the specific data properties of CNC machine tools, forming the standard GB/T 39561.4-2020 [40]. This paper establishes the corresponding IM of the CNC system according to the structure of the CNC equipment IM specified in the standard, describing its geometric parameters, physical attributes, relationships and characteristics, and other information. As shown in Fig. 6, the CNC system IM

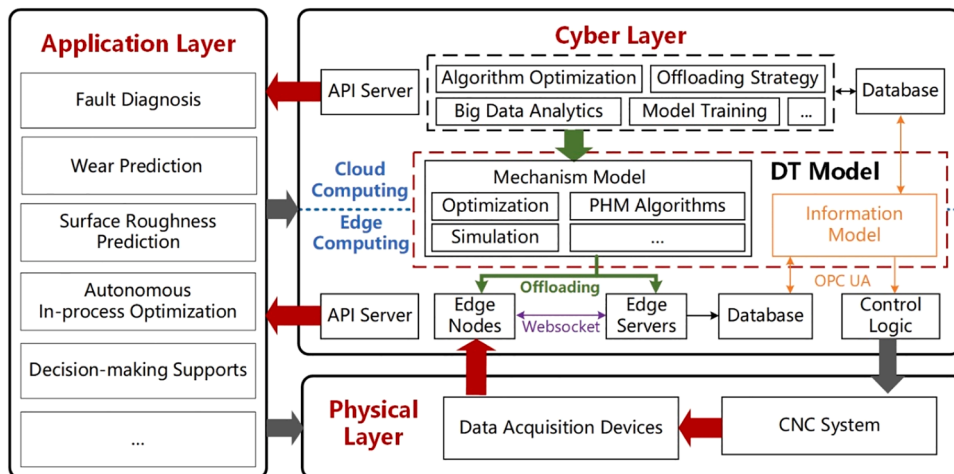


Fig. 4. Architecture of EI-driven DT of CNC system.

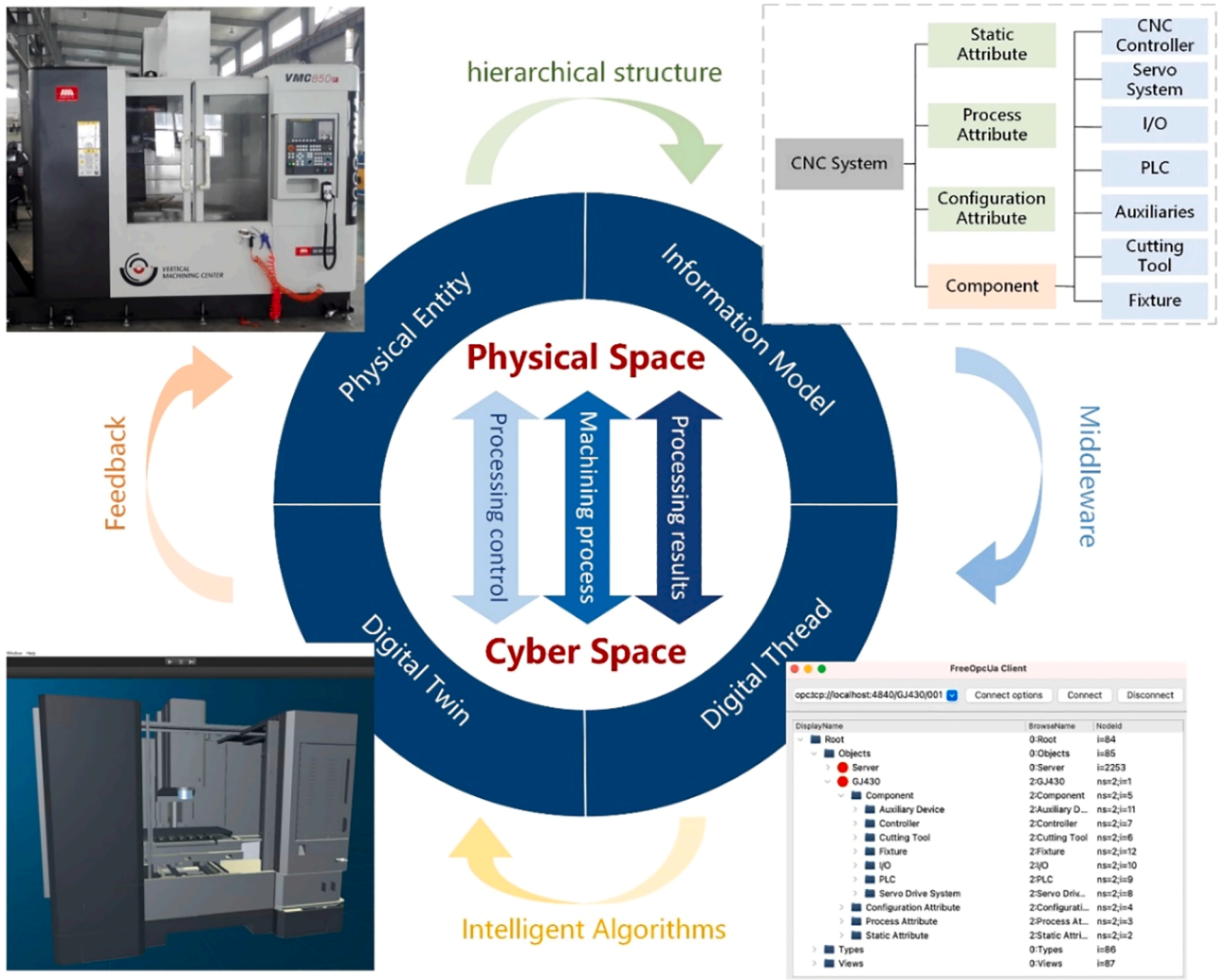


Fig. 5. DT modeling process.

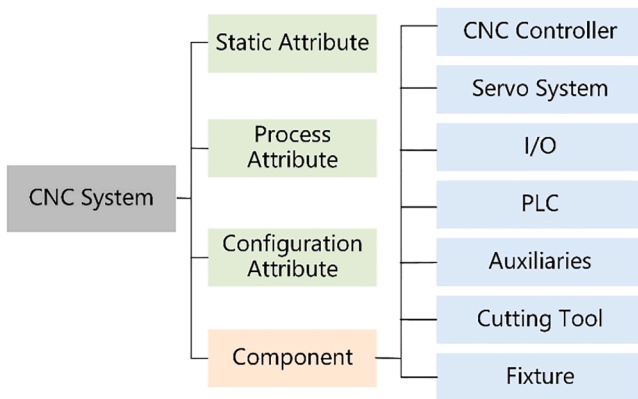


Fig. 6. The information model of CNC system.

consists of a set of static attributes, process attributes, configuration attributes, and a set of components. The static attribute set describes the inherent properties of the CNC system, the process attribute set describes the system's characteristics during operation, and the configuration attribute set is the relevant configuration required to perform a task. The component set is a physical or logical part of the CNC equipment description model and includes components such as CNC controller,

servo systems, I/O, PLC, auxiliaries, cutting tool and fixture.

Digital thread-Due to the diversity of CNC systems, communication is limited by the non-interoperability between different equipment suppliers' communication ports and protocols. It is not easy to achieve equipment integration and information sharing. Digital thread technology extends the scope and depth of data integration of DT, eliminates heterogeneity through template mapping, achieves consistent expression within typical standard specifications, and provides support for interconnection and interoperability between CNC equipment. The leading digital thread technologies used at the domestic and foreign are MTConnect and OPC UA (OLE for Process Control Unified Architecture). Compared to MTConnect, OPC UA adopts an object-oriented address space to establish the IM, supporting various data types such as attributes, characteristics, methods and events. OPC UA can dynamically adjust the IM by adding and removing nodes at server runtime, providing maximum convenience for users to customize the model [41, 42]. In order to advance the interaction and fusion between virtual and real data, this paper adopts the OPC UA approach to build the DT model with the middleware developed in previous research [43].

Mechanism model-Currently, most studies of DT model focus on building geometric and physical dimensions instead of MMs that can reflect the behavior and rules of physical entities [44]. Inadequate and incomplete DT models will undermine the realism of the virtual entities and, in turn, the accuracy of the simulation and prediction. Therefore, in addition to the IM, the DT model must also contain mechanistic models

that reflect the behavior of both the physical entities and the rules. On the one hand, this work develops AI algorithms-based theoretical derivations as a guide, combine them with empirical/control theory models to jointly build a comprehensive model with high fidelity and accuracy. On the one hand, this work develops AI algorithms-based theoretical derivations as a guide and combines them with empirical/control theory models to jointly build a comprehensive model with high fidelity and accuracy. On the other hand, considering that CNC system is a complex system whose behavior is influenced by multiple factors, multidisciplinary and multi-scale models are integrated to build a mechanistic model to ensure the integrity of the DT model.

Model Deployment & Optimization Decision-The IM and the MM can describe the properties, behaviors, and characteristics of the physical entity from different spatial and temporal perspectives, reflecting its operating state and changes in the machining process as time progresses. Together, they constitute a DT model that can better describe the real-time response and behavior of the CNC system in different environments, form optimal decision instructions, perform feedback control and guide the machining process [44,45]. To alleviate the pressure on the cloud, this paper deploys DT models distributed by EI based on the hierarchy of CNC systems. The intelligent algorithm is selected and trained in the cloud, the partitioning strategy of the task is developed, and the partitioned model is downloaded to the edge node and the edge server respectively, where the edge node and the edge server collaborate to process the real-time data.

3.4. EI based deployment method

Edge computing offers a low-latency approach compared to cloud computing. However, with the spread of IoT and AI, more and more sensors are deployed in industrial sites, the complexity of algorithms and the scale of data grows accordingly. In addition, DT applications such as real-time monitoring, fault diagnosis and prediction require more rigorous real-time data processing. The traditional data processing approach [46,47] uses a single server with single node serial execution to partition the data processing tasks, which is helpful to reduce latency. The tasks before the partition point are left to edge nodes, while the remaining tasks are offloaded to the edge server. The two parts of processing time are superimposed with the data transmission time as the data processing time. However, this data processing approach does not apply to DT of CNC system. On the one hand, the one-to-one correspondence between nodes and servers implies higher data processing costs. On the other hand, the latency of processing deep DNNs by edge servers is shorter. Moreover, this data processing approach, in which latency reduction is the only assessment metric, tends to give more data to edge servers for processing and ignore edge nodes. This method will lead to a large amount of data that the server is too late to process being discarded, and the system throughput rate will be lower, which helps less to the quality of the application service.

Considering the actual application scenario of DT of CNC system, this paper proposes a new method to partition tasks, which deploys edge nodes for each CNC system and deploys an edge server on the shop floor. This work uses the MISD (Multiple Instruction stream Single Data-stream) pipeline structure to process data between edge servers and edge nodes in a one-to-many relationship, in which the edge nodes are parallel while the edge server is serial. With τ as a timeslice, each processing level of the pipeline has dedicated functions and overlapping operations. This data processing method can make full use of the edge nodes, effectively reducing the amount of data discarded and improving the quality of application services. Details of the proposed method will be described as follows.

First, the data processing process is divided into N subtasks, S_i denotes the i -th subtask, where $i = 1, 2, \dots, N$, D_i denotes the size of the output data of S_i executed, and $f(S_i)$ is the time delay of S_i execution. The data processing task is partitioned by edge computing, and j denotes the number of subtasks divided to be processed by the edge node, where $j =$

$0, 1, \dots, N$. When $j = 0$, all subtasks are offloaded to the edge server for execution, and when $j = N$, all subtasks are executed at the edge node.

In the next step, the latency model for each segment of the MISD pipeline is given. The data processing delay consists of edge node execution, network transmission, and edge server execution.

1) Edge node tasks model

Edge nodes process multi-group data in parallel batches. Assuming that each edge node receives the same data processing task and the nodes have the same computing power, for the data processing task of any node, the execution latency of the edge node is divided into two parts: the node processing sub-task latency before the split point and the result upload latency. Defining the currently available bandwidth as B_{device} , the edge node execution latency is noted as

$$t_{device} = \sum_{i=0}^j f(S_i) + D_j / B_{device} \quad (1)$$

1) Edge server tasks model

The currently available bandwidth of the edge server is denoted as B_{edge} , and the execution latency when connecting to a single node includes subtask processing latency and data download latency, denoted as

$$t'_{edge} = \sum_{i=j+1}^N f(S_i) + D_j / B_{edge} \quad (2)$$

The actual number of nodes accessed by the edge server is recorded as m , and the maximum number of accessible nodes is recorded as M , where $m \leq M$. Nonpreemptive CPU allocation is used for serial execution, and idle time may occur. Consequently, the actual processing latency t_{edge} of the edge server should satisfy $mt'_{edge} \leq t_{edge} \leq Mt'_{edge}$.

1) Communication model

The data communication latency t_{tran} consists of data propagation latency and forwarding latency. Among them, propagation latency is the time required for data to propagate over the transmission channel and is only related to the transmission channel length, while forwarding latency contains the queuing delay of data in the input and output buffers and the processing latency required to perform forwarding to process the data blocks.

Finally, the task slicing criteria are given. The maximum rate of outputting one result per clock cycle can be guaranteed only in a steady-state where the pipeline is loaded [48]. Therefore, the time required for the execution of the slowest component in the 3-stage pipeline is the timeslice length, i.e., $\tau = \max\{t_{device}, t_{tran}, t_{edge}\}$. The pipeline processing each process needs to wait for the data processed by the previous process, and the data processing delay is $T = t_{device} + t_{tran} + t_{edge}$. When the pipeline is full, the system throughput rate $\theta = 1/\tau$, the smaller the selected timeslice is, the larger the system throughput rate will be if the time delay requirement is satisfied. The high throughput rate means fewer data discarded, which is beneficial for model accuracy.

4. Task deployment algorithm for EI-driven DT

4.1. Task partitioning algorithm

Section 3.4 uses edge computing to partition and deploy intelligent algorithms in the hope of maximizing the system throughput rate and further improving system's applications efficiency while ensuring real-time performance. The optimal partition point for the model depends on the DNN's topology, computational capacity of the device, and the

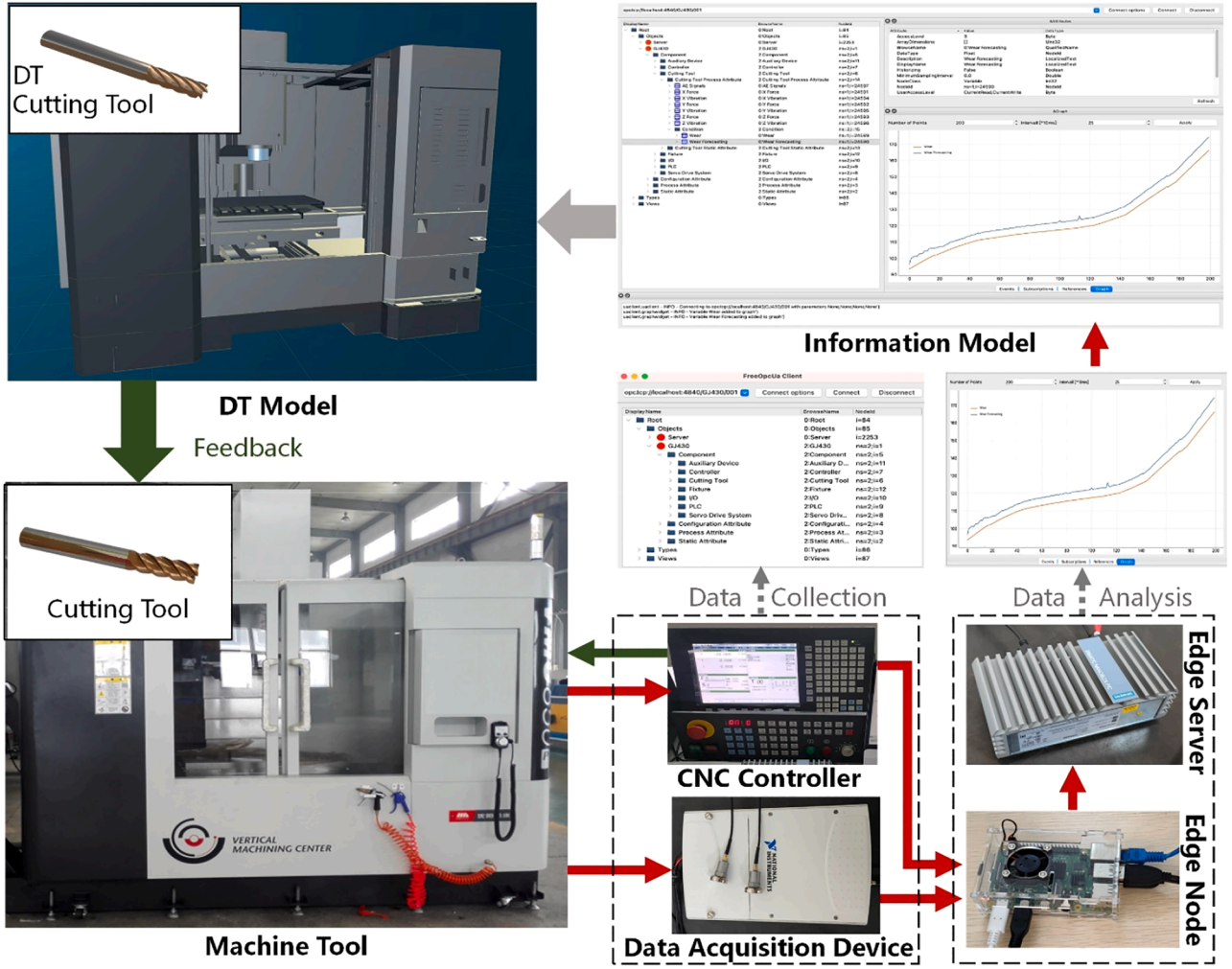


Fig. 7. The experimental environment and the workflow.

network state, which manifests itself in computation and data size variations of each subtask. The specific task partitioning algorithm is as follows.

Task initialization—Obtain the computational capacity of edge nodes and the edge server and analyze the target subtasks. For the data processing task that contains N subtasks, the execution latency TN_i at the edge node and the latency TE_i at the edge server is initialized for each subtask S_i , respectively. Since the inference latency of subtasks depends only on the computational capacity of the hardware device, the same hardware needs to be initialized only once for the same hardware, and the latency of the device to perform each subtask $f(S_i)$ can be obtained from actual measurements.

Task Analysis—According to the description in Section 3.4, the edge nodes execute the subtasks before the PartitionPoint in parallel. The data processing time is $\sum_{k=0}^j TN_k$, and the upload time is D_i / B_{device} . The edge server serially executes the remaining subtasks that each node offloads to the server. Since the nodes send tasks to the edge server at uncertain times, the edge server may be idle. Moreover, mark the task execution time of the edge server as $M(\sum_{k=j+1}^N TE_k + D_j / B_{edge})$. Due to the complicated calculation of data transmission time t_{tran} , this paper uses real-time monitoring to obtain it.

PartitionPoint Selection—This paper uses the timeslice length to describe the system throughput rate. Furthermore, the latency and timeslice length are calculated corresponding to all candidate points. The candidate points are after each subtask. For the candidate points that meet the latency requirements, this paper compares their timeslice

lengths one by one. The one with the most petite timeslice length, i.e., the most extensive system throughput rate, is selected as the optimal PartitionPoint. The partition decision and corresponding system latency are output.

4.2. Model selection algorithm

Algorithm 1 can select the optimal PartitionPoint for data processing tasks and maximize the throughput rate based on latency requirements for a given algorithm model. Nevertheless, applications have different considerations for throughput rate and model accuracy in industrial sites. Methods such as model pruning can effectively increase the throughput rate at the cost of model accuracy. So, it becomes a new problem to balance model accuracy and throughput rate and select the appropriate algorithm model for different applications.

This section proposes a method for adaptive selection of algorithmic models based on Algorithm 1. It provides an automated and timely evaluation of different models for applications that utilize an evaluation function based on the system latency, accuracy, and throughput of alternative algorithmic models.

The EI-driven DT mainly applies machine learning to solve classification and regression problems. For both types of problems, different metrics are used to evaluate the capability of their algorithms. This paper defines the evaluation functions are defined for them, respectively.

For the classification problem, the evaluation function is defined as

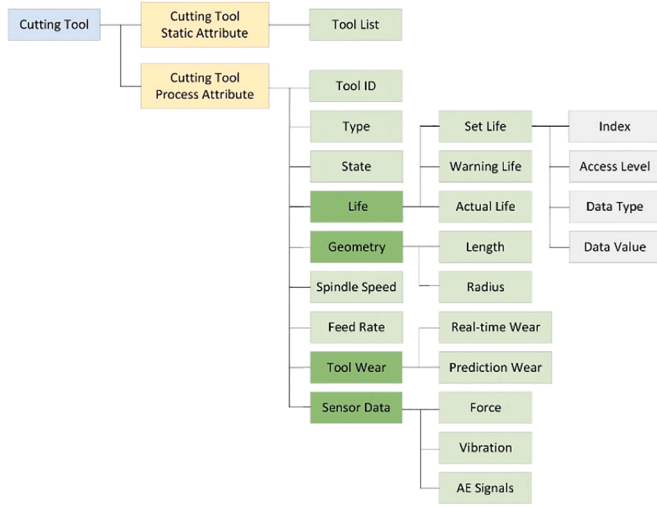


Fig. 8. Information model of cutting tool.

follows.

$$h(acc, \theta, p) = (l\theta/p)^{a \cdot (acc - Y)} \quad (3)$$

where acc is the accuracy rate of the DNN model, l is the latency requirement, θ is the system throughput rate, p is the number of pipeline segments, Y is the accuracy threshold, and a is a constant factor greater than 0, which user can set according to production requirements.

For the regression problem, the fit error of the model is denoted by $loss$, and the evaluation function is as follows.

$$h(loss, \theta, p) = (l\theta/p)^{a \cdot (Y - loss)} \quad (4)$$

Its specific selection method is the same as the classification problem, with the evaluation function calculated from Eq. (4). The smaller the fitting error of the alternative model, the higher the evaluation value. The alternative model with the highest evaluation value is output.

The DNN model adaptive selection algorithm is shown in Algorithm 2. First, input alternative DNN models with the maximum tolerable delay and accuracy requirements of the application. Initialize the throughput rate, accuracy rate, or error of the alternative DNN models. Next, calculate the evaluation value $E(M_j)$ of all alternative DNN models. Finally, the one with the highest evaluation value is selected as the optimal DNN model output.

5. Experimental study

CNC milling machines are widely used in the field of machine building due to their high production efficiency. As the direct executor of cutting, the tool is the most easily damaged component of a CNC milling machine. The traditional cutting process mainly relies on the subjective experience of the processing personnel that judge the cutting tool wear to change the tool. This can result in instances where a tool is withdrawn before it wears out beyond its limitations, or where a tool is worn out violently but not replaced [49]. Underused tools will lead to higher production costs. At the same time, failure to change tools on time will affect product quality and machining accuracy, which damages the machine tool and endangers personnel safety. Therefore, real-time monitoring and prediction of cutting tool wear are needed.

To address the above issues, this section gives a case of EI-driven DT of CNC system, establishes a DT model of the cutting tool, disposes the edge computing environment, and deploys the DT model through the task partitioning and model selection algorithms proposed in this paper.

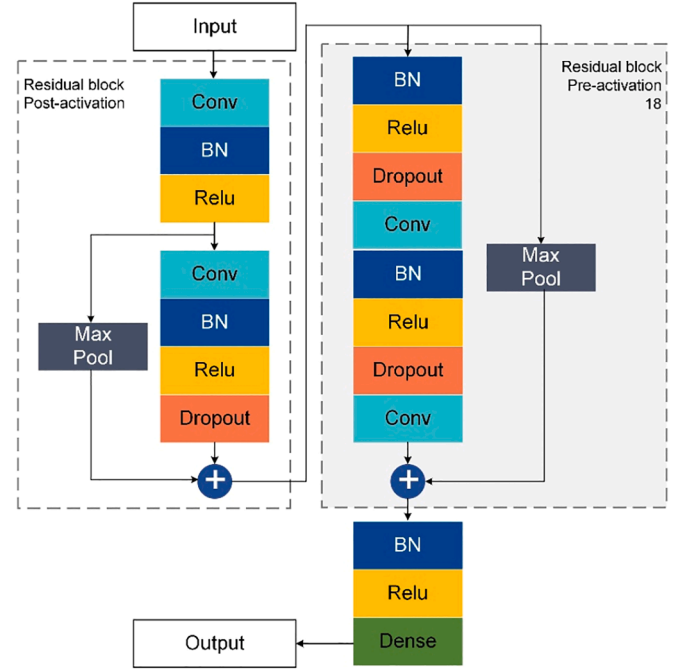


Fig. 9. The real-time monitoring model of tool wear based on the ResNet.

5.1. Experimental setup

Fig. 7 shows the experimental environment and workflow in which the CNC controller adopts Golding CNC system GJ430, the edge node adopts RaspberryPi 4B, and the edge server adopts SIMATIC NANOBOX PC. GJ430 and the data acquisition device collect real-time data generated during processing to build the IM. The edge nodes and edge servers are deployed with intelligent algorithms to diagnose and predict tool wear. Finally, this work monitors the tool status in real time through the visualization client and provides feedback control through GJ430.

5.2. Information modeling of cutting tools

This section establishes an IM of the cutting tool according to the method given in section 3.3, as shown in Fig. 8. The cutting tool is a crucial component of the CNC system, which contains the process attribute set and static attribute set. The data of the static attribute set is the tool list. The process attributes set includes the information of the tool used, such as tool ID, type, state, life (set life, warning life and actual life), the geometric parameters (tool's length and radius), the rated spindle speed and feed rate, the tool wear (actual and predicted wear) and the sensor data, including mechanical, acceleration and Acoustic Emission (AE) signals. Each data item contains a description of the index, access level, data type and data value.

The mapping process from the IM to the DT model is implemented using OPC UA. Each data item in the IM corresponds to a node in the OPC UA address space, and the reference relationship between the nodes is determined by the hierarchy of the IM. Users can access the server through various clients supporting the OPC UA standard to obtain the tool information in the OPC UA address space to load, browse and modify the tool IM.

Tool management and maintenance can be achieved by subscribing to the OPC UA server node "Wear Prediction". When the wear prediction value reaches the warning life, the OPC UA server can send the fault information to the CNC system in real-time through the error channel provided by the CNC system, which can trigger the alarm and display the corresponding tool fault information on the HMI of the CNC system, which can prompt the operator to manage and maintain the tool.

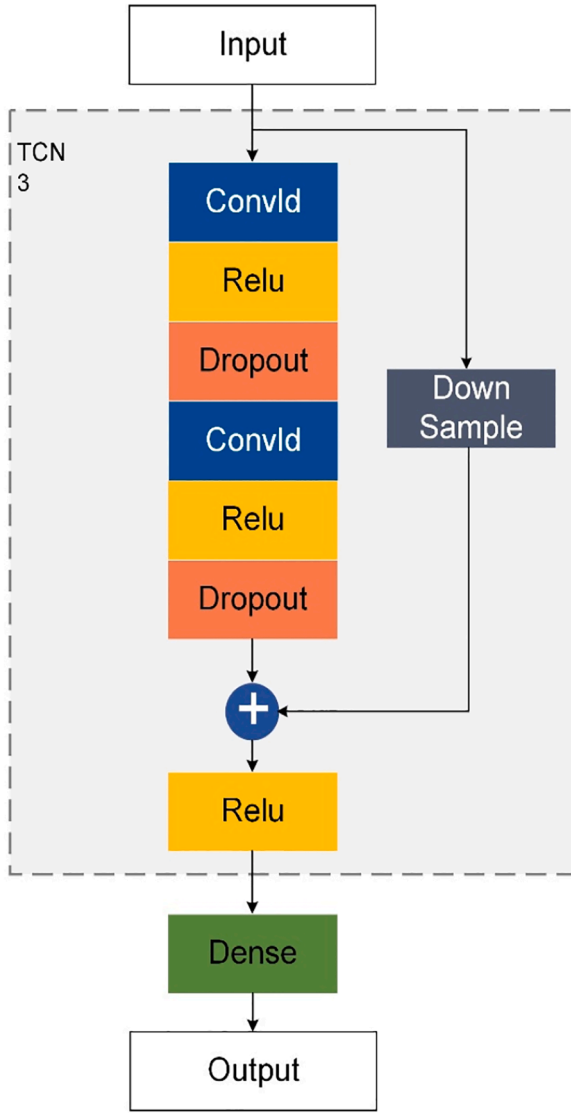


Fig. 10. TCN-based tool wear forecasting model.

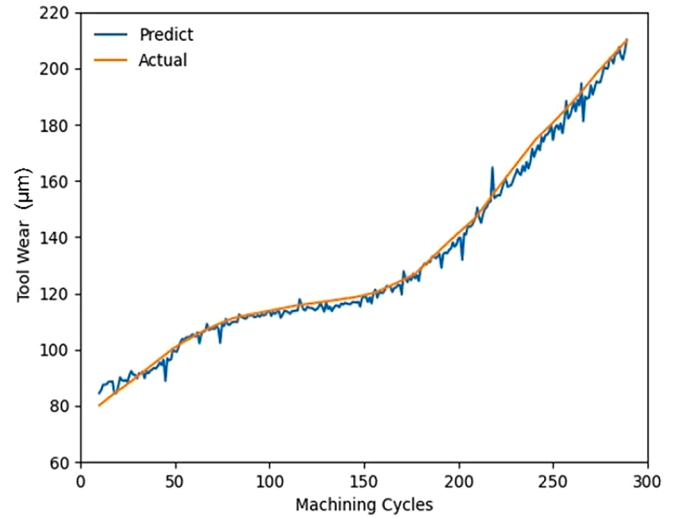
5.3. Mechanism model deployment

5.3.1. Model selection

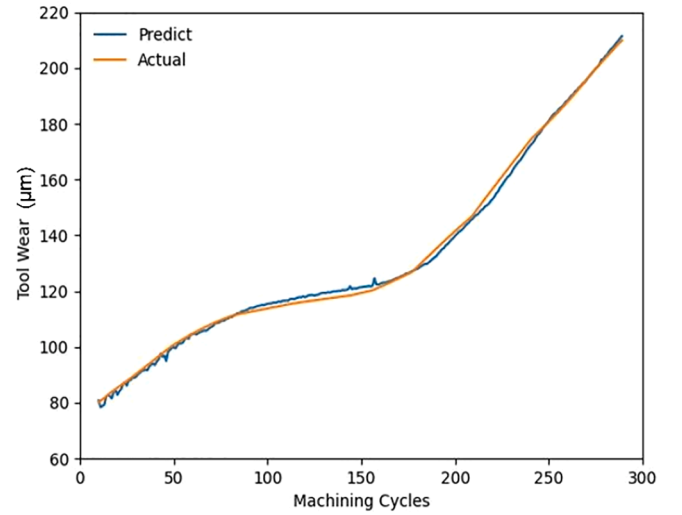
Tool wear is a dynamic, time-varying, non-linear, and stochastic process with many possible interferences. DNN has good learning abilities and can establish complex relationships between sensor signals and actual tool wear through multi-layer representations. Existing studies generally agree that the deeper the network is, the more accurate the model is. Nevertheless, degradation often occurs along with the increase of network depth. The ResNet can effectively solve the deep network degradation phenomenon while guaranteeing network depth. The literature [50] used ResNet to monitor tool wear in real-time. The real-time monitoring model of tool wear based on the ResNet is shown in Fig. 9, in which Conv represents the convolutional layer, BN represents the batch normalization layer, and Dense represents the fully connected layer.

On this basis, Sun et al. [50] used LSTM to predict the tool wear condition. However, Temporal Convolutional Network (TCN) has a stable gradient, fast training speed, flexible acceptance domain size, and variable input length [51]. To select a suitable prediction model, a 3-block TCN model, as shown in Fig. 10, is designed in this paper.

The existing LSTM model and the TCN model designed in this paper were trained for 200 cycles, respectively. The input data size of the



(a) LSTM (d=15, s=10)



(b) TCN (d=15, s=10)

Fig. 11. The results of the tool wear prediction.

Table 1

Model Comparison

DNN Model	RMSE	MAE	Node execution delay (ms)	Server execution delay (ms)
LSTM	7.47	6.21	12.27	6.82
TCN	3.67	3.17	20.45	11.37

model was the tool wear value after 15 machining sessions, and the prediction step was 10 machining sessions. The results of the tool wear prediction are shown in Fig. 11. The RMSE (Root Mean Squared Error) and MAE (Mean Absolute Error) were used to describe the model loss. As shown in Table 1, the TCN model has better real-time performance with a minor deviation in the model loss. According to Algorithm 2, the TCN model is finally selected for tool wear state prediction in this paper.

5.3.2. Task partition

This paper uses ResNet and TCN to diagnose and predict tool wear conditions. The data processing flow is shown in Fig. 12. The data processing tasks are partitioned and deployed according to Algorithm 1, where the candidate points are located after each subtask. The subtasks are divided as shown in Table 2. The input data are 7-dimensional time series consisting of X-axis force, Y-axis force, Z-axis force, X-axis

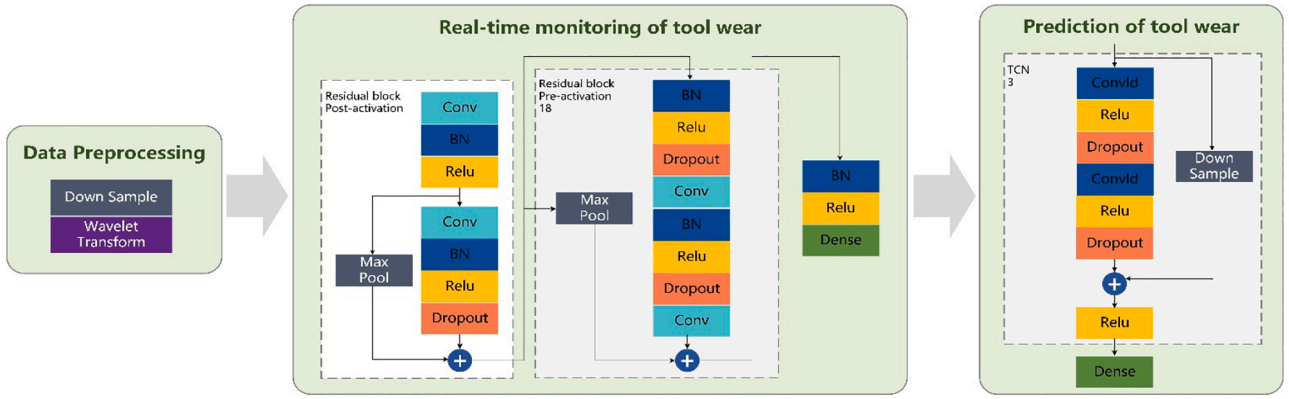


Fig. 12. The data processing flow.

Table 2
Subtask division

stage	Subtask	Component	output
Data pre-processing	0	Down sampling	7*5000
		Wavelet transform	7*1255
Real-time tool wear monitoring	1	Residual block Post-activation	128*626
	2	Residual block Pre-activation [0]	128*313
	3	Residual block Pre-activation [1]	128*313
	4	Residual block Pre-activation [2]	128*157
	5	Residual block Pre-activation [3]	128*157
	6	Residual block Pre-activation [4]	128*79
	7	Residual block Pre-activation [5]	128*79
	8	Residual block Pre-activation [6]	128*40
	9	Residual block Pre-activation [7]	128*40
	10	Residual block Pre-activation [8]	128*20
	11	Residual block Pre-activation [9]	128*20
	12	Residual block Pre-activation [10]	128*10
	13	Residual block Pre-activation [11]	128*10
	14	Residual block Pre-activation [12]	128*5
	15	Residual block Pre-activation [13]	128*5
	16	Residual block Pre-activation [14]	128*3
	17	Residual block Pre-activation [15]	128*3
	18	Residual block Pre-activation [16]	128*2
	19	Residual block Pre-activation [17]	128*2
Tool wear forecasting	20	Dense	1*1
	21	TCN[0]	4*15
	22	TCN[1]	8*15
	23	TCN[2]	16*15
	24	Dense	1*15

vibration, Y-axis vibration, Z-axis vibration, and AE signals collected by sensors and GJ430. The specific task partitioning steps are as follows.

First, initialize the execution latency of each subtask at the edge node and server. The record is shown in Fig. 13.

Next, calculate the data processing latency and throughput rate of

Algorithm 1
Task Partitioning Algorithm

```

1: Input:
2: N: number of the subtasks
3:  $\{S_i | i = 1, \dots, N\}$ : the  $i$ -th subtask
4:  $\{D_i | i = 1, \dots, N\}$ : output data size of  $S_i$ 
5:  $f(S_i)$ : latency of executing  $S_i$ 
6:  $B_{device}$ : current available bandwidth of node
7:  $B_{edge}$ : current available bandwidth of edge server
8: M: maximum number of nodes that can connect to the server
9: l: latency requirement
10: Output:
11: Partition Decision
12: procedure
13: for each  $i$  in  $1, \dots, N$  do
14:    $TN_i \leftarrow f_{device}(S_i)$ 
15:    $TE_i \leftarrow f_{edge}(S_i)$ 
16: end for
17:  $\tau_{min} \leftarrow l$ , PartitionPoint  $\leftarrow 0$ 
18: for  $j = 0, j \leq N, j = j + 1$  do
19:    $t_{device} = \sum_{k=0}^j TN_k + D_j / B_{device}$ ,  $t_1 \leftarrow t_{device}$ 
20:    $t_2 \leftarrow t_{tran}$ 
21:    $t'_{edge} = \sum_{k=j+1}^N TE_k + D_j / B_{edge}$ ,  $t_3 \leftarrow M'_{edge}$ 
22:    $\tau_j = \max\{t_1, t_2, t_3\}$ ,  $T_j = t_1 + t_2 + t_3$ 
23:   if  $T_j \leq l$  then
24:     if  $\tau_j \leq \tau_{min}$  then
25:        $\tau_{min} \leftarrow \tau_j$ , PartitionPoint  $\leftarrow j$ 
26:   end if
27: end if
28: end for
29: return PartitionPoint and  $T_{min}$ 

```

each candidate point. The optimal PartitionPoint is defined by two metrics: inference latency and throughput rate. As shown in Fig. 14, this paper uses different colors to indicate the processing delay of the data at different stages of the pipeline and uses a different symbol to differentiate among the different metrics used in making the decision. The diamond symbol marks the partition point with the highest throughput rate. The star symbol marks the partition point with the highest throughput rate when the inference delay meets the real-time requirement, i.e., the optimal PartitionPoint. The maximum tolerable latency of the system is set at 480ms. When the edge server connects two edge nodes (Fig. 14 (a)), the inference latency of the PartitionPoint corresponding to the optimal throughput rate satisfies the latency requirement, which is the optimal partition point. When the server is connected to four edge nodes (Fig. 14 (b)), the inference delay of the partition point corresponding to the optimal throughput rate exceeds the maximum tolerable latency. Furthermore, the candidate points with the highest throughput rate among those meeting the latency requirement are selected as the optimal PartitionPoint for model partitioning.

Algorithm 2**Model Selection Algorithm**

```

1: Input:
2:  $\{M_j | j = 1, 2, \dots, R\}$ : the DNN model
3:  $l$ : latency requirement
4:  $Y$ : threshold
5:  $\alpha$ : custom constant coefficient
6:  $E(M_j)$ : the evaluation of DNN model  $M_j$ 
7: Output:
8: Optimal DNN Model
9: Procedure
10: for ( $j = 1; j \leq R; j++$ ) do
11:  $h_{max} = 0, M_{optimal} = 0$ 
12: if  $Target == accuracy$  then
13:  $h_j(acc_j, \theta_j, p_j) \leftarrow E(M_j)$ 
14: else if  $Target == loss$  then
15:  $h_j(loss_j, \theta_j, p_j) \leftarrow E(M_j)$ 
16: end if
17: if  $h_{max} \leq E(M_j)$  then
18:  $h_{max} = E(M_j), M_{optimal} = M_j$ 
19: end if
20: end for
21: return the optimal DNN model  $M_j$ 

```

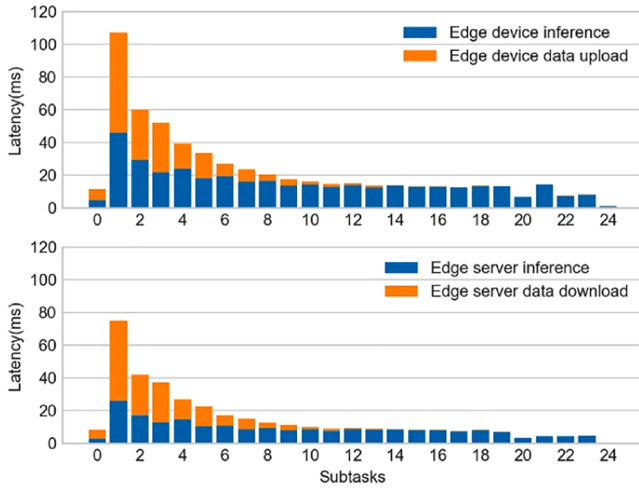


Fig. 13. The execution delay of subtasks.

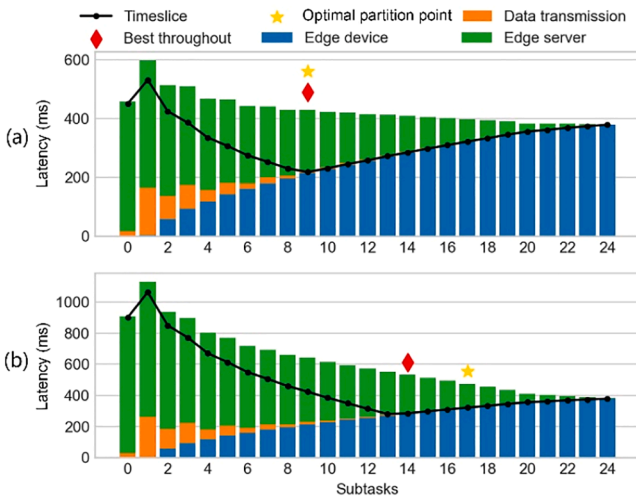


Fig. 14. The PartitionPoint selection.

5.4. Digital twin application

The edge node executes the model before the PartitionPoint and sends the result to the edge server via WebSocket. The edge server executes the model after the PartitionPoint and sends the result to the OPC UA server. The user obtains the tool condition by accessing the server through the client. OPC UA offers two main communication methods: Client-Server and Publish-Subscribe (PubSub). The server provides the client's access information through defined services [52], while PubSub allows multiple authorized clients (subscribers) on the LAN (Local Area Networks) to access data published by the OPC UA server (publisher).

This paper develops an OPC UA server based on the CNC system IM and integrates the server into the GJ430 CNC system. The server configuration, such as port number and security certificate, is set in the server initialization phase. Then the node attributes and inter-node reference relationships are defined in the OPC UA server-side node space according to the CNC system IM provided by the XML file, establishing the corresponding OPC UA model on the server-side. Bind the data sources of the data nodes to the data reading functions provided by the GJ430 CNC dynamic link library to achieve real-time updates of the node data. Finally, the server listener is started, and the access address of the server address space is returned in the console interface.

In order to realize the diagnosis and prediction of tool wear, the OPC UA server sets the *Warning Life* as the trigger condition of the event. The event is triggered when the server detects that the *Prediction Wear* has reached its *Warning Life*, i.e., tool wear is close to exceeding the normal range, the event is triggered. Moreover, all the clients subscribed to the event can get the notification information from the server. Furthermore, the field operators are prompt to carry out cutting tool maintenance.

Any client that supports the OPC UA standard can access the DT model of the CNC system via the port provided by the OPC UA server. The implementation of DT is not client-dependent, and the user has more flexibility in applying the DT model. This paper, therefore, uses an open-source client interface to present tool wear and prediction results.

As shown in Fig. 15, the client contains three main areas. To connect the client with the server, enter the URL in the address bar at the top. The client connection status is displayed at the bottom of the screen. After successful connection, area 1 displays the currently connected CNC system (GJ430)'s IM. The OPC UA model nodes provided by the server are displayed in a tree structure. Users can browse the IM to understand the logical structure of the CNC system and the data items available for access. By clicking on a data item in the node list, users can view the attributes of the selected node in area 2, including *AccessLevel*, *NodeId*, *NodeClass* and *Data Type*. Through the analysis of real-time data, the invisible variation is made explicit in area 3, including subscription event alerts, subscription data item display, server reference relationship, and image display. As shown in Fig. 15, the 'Prediction Wear' and 'Real-time Wear' data items are loaded into the graph block. Set the sampling interval as 250ms, the change curve of the tool wear condition can be viewed in real-time. The horizontal coordinate represents the time, while the vertical coordinate represents the wear value (μm).

This case of EI-based DT allows users to obtain real-time tool status quickly and reliably. Based on this, operators can carry out predictive maintenance of tools, which can improve the machining quality, further ensure remaining tool life utilization to reduce the production cost.

6. Conclusion

The DT of CNC system is one of the most promising technologies to realize intelligent manufacturing. This paper starting from the development needs of intelligent manufacturing, develops the research on the critical technology of DT of CNC system based on edge intelligence, and puts forward the solution of DT of CNC system in view of the limited amount of computation and network resources in industrial field, which cannot meet the requirements of intelligence and real-time. The main contributions are summarized below.

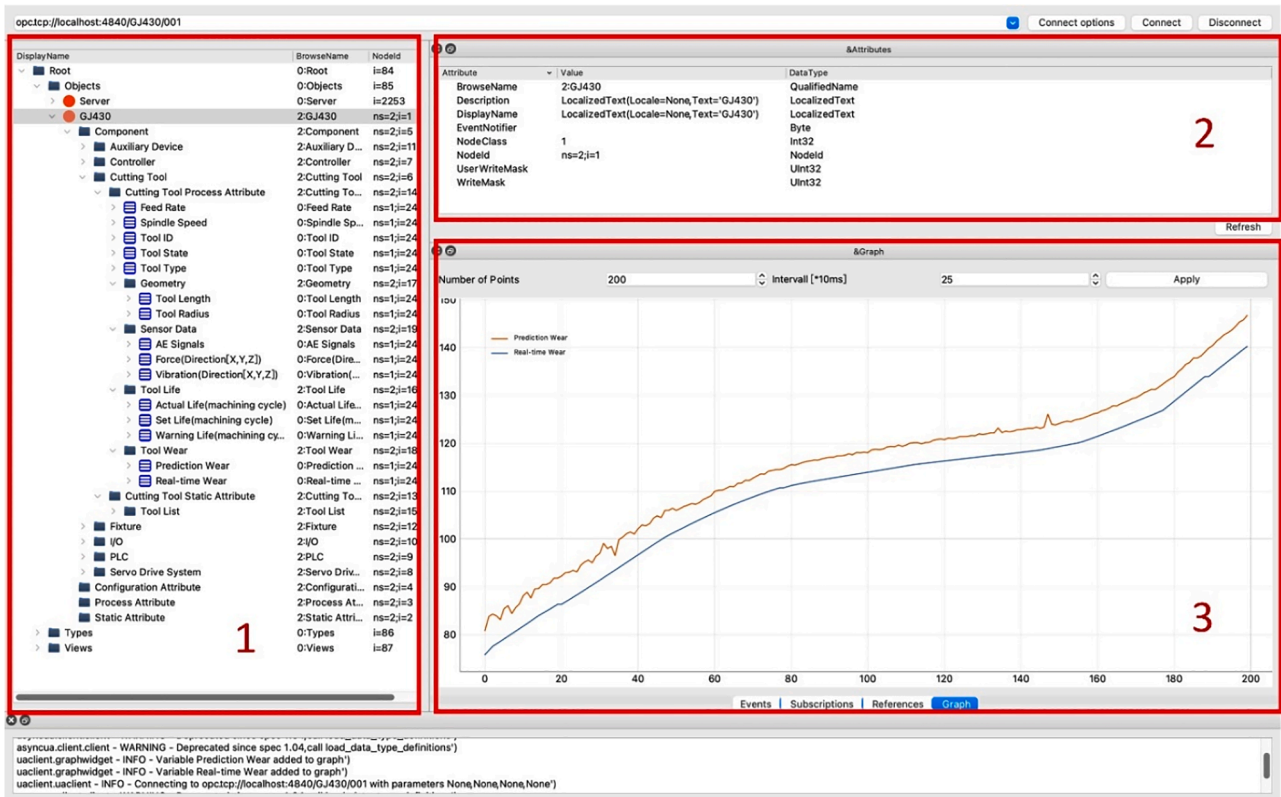


Fig. 15. The visualization client.

First, this paper analyzes the demands of the intelligent CNC system and proposes a theoretical method of hierarchical modeling for the CNC system to break through the bottleneck of real-time service with edge computing. In this context, this paper proposes a DT architecture of the CNC system driven by EI, and the DT modeling method of the CNC system is described in detail. Besides, in the distributed deployment process of hierarchical DT models of the CNC system, a model partitioning algorithm is proposed to improve the model throughput rate based on the guaranteed real-time service. In order to further reduce the manual overhead of model profiling at the install phase, this paper proposes a model selection algorithm that provides timely automatic model evaluation for applications. The various requirements on throughput rate and model accuracy are satisfied. Finally, a case study of tool wear diagnosis and prediction through the EI-driven DT during the machining process is provided.

Realizing the intelligence of CNC systems is the main objective. The research work presented in this paper only provides a preliminary discussion on the construction method and deployment method of the DT model of CNC systems, and the proposed EI-driven DT architecture of CNC systems still needs to be enriched and improved. In terms of future research, we will provide more functions and services for users to enrich its mechanistic model and expand the application of the proposed method to practical functions such as trajectory planning, surface roughness inspection and tool chattering. And this paper assumes that all edge nodes have the same task model when deploying DT models, the next step is to study how to partition the deployment of intelligent algorithm models when the edge nodes have different task models in the future according to the actual situation of the shop floor.

CRedit authorship contribution statement

Haoyu Yu: Conceptualization, Methodology, Software, Formal analysis. **Dong Yu:** Supervision, Project administration. **Chuting Wang:** Resources, Data curation, Writing – original draft. **Yi Hu:** . **Yue Li:** .

Declaration of Competing Interest

All authors disclosed no relevant relationships.

Data Availability

Data will be made available on request.

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